Empirical Scaling Laws for Neural Language Models (Kaplan et al. 2020)

Language model cross-entropy loss scales as a power-law with compute, dataset size, and parameter count across more than seven orders of magnitude, with architectural hyperparameters having minimal effect.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL, ORGANIZATIONAL, PHILOSOPHICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Cross-entropy loss in autoregressive Transformer language models follows predictable power-law relationships with three fundamental variables: parameter count N, dataset size D, and compute budget C. Architectural details such as network depth, width, and number of attention heads have remarkably little impact within wide bounds, establishing that scale dominates structural tuning.

EVIDENCE AND CAUSAL ASSESSMENT

Systematic ablation across model sizes from 10^3 to 10^9 parameters and compute budgets demonstrated smooth, continuous power-law decay of test loss without plateauing, indicating that language modeling difficulty scales predictably with information-theoretic information density.

EFFECT SIZE AND PERSISTENCE

Performance improves smoothly as L(N) ≈ (N_c/N)^{α_N}, with power-law exponents α_N ≈ 0.076, α_C ≈ 0.050, and α_D ≈ 0.095. Models can be scaled reliably by predicting loss before committing massive training clusters.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as: robust (<20%).

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WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Power-law reduction in next-token cross-entropy loss does not guarantee semantic comprehension, factual reliability, or causal reasoning ability in downstream inference.

WHAT ACTION IS REASONABLY JUSTIFIED

Use scaling law equations to determine optimal compute and data allocations prior to large-scale training runs rather than heuristic trial-and-error parameter tuning.

VERIFIED SOURCES & DOI IDENTIFIERS

- Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., & Amodei, D. (2020). Scaling Laws for Neural Language Models. arXiv preprint arXiv:2001.08361. https://arxiv.org/abs/2001.08361